

Communication protocol

The instrument provides a standard RS485 communication interface and adopts the MODBUS-RTU communication protocol.

Types and formats of communication data

The information transmission is asynchronous and in byte units. The communication information transmitted between the host and slave is in 11 bit word format, including 1 start bit (0), 8 data bits, and 2 stop bits (1). The format of the information frame is as follows:

Start	Address code	Function code	Data area	CRC check	Finish
Pause time greater than 3.5 characters	1 byte	1 byte	n byte	2 byte	Pause time greater than 3.5 characters

Communication information transmission process

When a communication command is sent from the host to the slave, the slave that matches the address code sent by the host receives the communication command. If the CRC check is correct, the corresponding operation is performed, and then the execution result (data) is returned to the host. The returned information includes the address code, function code, executed data, and CRC check code. If the CRC check fails, no information will be returned.

Address code

The address code is the first byte of each communication information frame, ranging from 1 to 247. Each slave must have a unique address code within the bus, and only the slave that matches the address code sent by the host can respond to the echo message. When the slave sends back information, the returned data starts with their respective address codes. The address code sent by the host indicates the address of the slave to be sent, the address code returned by the slave indicates the address of the slave to be sent back, and the corresponding address code indicates where the information comes from.

Function code

The second byte of each communication information frame. The host sends a function code to tell the slave what action should be performed. The slave responds by returning the same function code as the one sent from the host, indicating that the slave has responded to the host and performed the relevant operation. The instrument supports the following two function codes:

Function code (hexadecimal)	Definition	Operation
03H	Read Register	Read data from one or more registers
10H	Write multiple registers	Write n 16 bit binary data into n consecutive registers

Data area

The data area varies depending on the function code. These data can be numerical values, reference addresses, etc. For different slaves, the address and data information are not the same, and a communication information table should be provided.

Introduction of Function Codes

Function code 03H: Read register

For example, if the host wants to read data from two registers with a slave address of 01H and a starting register address of 0CH, the host sends:

Host sends		Sent information
Address code		01H
Function code		03H
Starting register address	High Byte	00H
	Low Byte	0CH
Number of Registers	High Byte	00H
	Low Byte	02H
CRC check	Low Byte	04H
	High Byte	08H

If the data in the slave registers 0CH and 0DH are 0000H and 1388H, the slave returns:

Return from slave		Returned information
Address code		01H
Function code		03H
number of bytes		04H
Register 0CH data	High Byte	00H
	Low Byte	00H
Register 0DH data	High Byte	13H
	Low Byte	88H
CRC check code	Low Byte	F7H
	High Byte	65H

Function code 10H: Write multiple registers

For example, if the host wants to save data 0002H, 1388H, and 000AH to three registers with slave address 01H and starting register address 00H, the host sends:

Host sends		Sent information
Address code		01H
Function code		10H
Starting register address	High Byte	00H
	Low Byte	00H
Number of Registers	High Byte	00H
	Low Byte	03H
Number of Registers which be wrote		06H
00H Register waiting to write data	High Byte	00H
	Low Byte	02H
01H Register waiting to write data	High Byte	13H
	Low Byte	88H
02H Register waiting to write data	High Byte	00H
	Low Byte	0AH

CRC check code	Low Byte	9BH
	High Byte	E9H

Return from slave:

Return from slave		Returned information
Address code		01H
Function code		10H
Starting register address	High Byte	00H
	Low Byte	00H
Number of Registers	High Byte	00H
	Low Byte	03H
CRC check code	Low Byte	80H
	High Byte	08H

16 bit CRC check

The host or slave can use a check to determine whether the received information is correct. Due to electronic noise or other interference, errors may occur during the transmission of information. The check can verify whether the communication information between the host or slave is incorrect.

The 16 bit CRC checks is calculated by the host and placed at the end of the transmitted information frame. Re calculate the CRC of the received information from the slave, compare the calculated CRC with the received CRC to see if they are consistent. If they are not consistent, it indicates an error. Only 8 data bits are used for CRC calculation, and the start and stop bits do not participate in CRC calculation.

The calculation method of CRC check code is as follows:

- ① Pre set one 16 bit register as hexadecimal FFFF (i.e. all 1s), referred to as the CRC register;
- ② Put the first 8-bit binary data (the first byte of the communication information frame) that is different from the lower 8 bits of the 16 bit CRC register, and place the result in the CRC register;
- ③ Move the content of the CRC register to the right by one bit (towards the lower bit)

and fill the highest bit with 0, and check the shifted bit after the right shift;

④ If the displacement is 0: repeat step ③ (move one more position to the right);

If the offset is 1: XOR the CRC register with polynomial A001 (1010 0000 0000 0001);

⑤ Repeat steps ③ and ④ until the entire 8-bit data is processed by shifting to the right 8 times;

⑥ Repeat steps 2 to 5 to process the next byte of the communication information frame;

⑦ After calculating all bytes (excluding CRC checksum) of the communication information frame according to the above steps, the content of the CRC register obtained is the 16 bit CRC check.

Error Handling

When the instrument detects errors other than CRC check code errors, it will send a message back to the host. The highest position of the function code is 1, which means that the function code returned from the slave to the host is based on the function code sent by the host plus 128. The error message frame format returned by the slave is as follows:

Address code	Function code (highest digit is 1)	error code	CRC check low byte	CRC check high byte
1 byte	1 byte	1 byte	1 byte	1 byte

The error code is as follows:

01H	Illegal function code	The received function code is not supported by the controller
02H	Illegal register address	The received register address exceeds the register address range of the instrument
03H	Illegal number of registers	The number of received registers exceeds the allowed number of registers
04H	Illegal data values	The received data value exceeds the data range of the corresponding address

Communication Information Table

Address	Parameter code	Parameter Name	read-write property	remarks
00H	C 01	The set value of the first segment of the first curve	R/W	
01H	t 01	Time setting for the first segment of the first curve	R/W	
02H-27H	C02-t20	Setting of the 2nd to 20th segments of the first curve	R/W	
28H	C 01	Set value for the first segment of the second curve	R/W	
29H	t 01	Time setting for the first segment of the second curve	R/W	
2AH-4FH	C02-t20	Setting of the 2nd to 20th segments of the 2nd curve	R/W	
50H	C 01	The set value for the first segment of the third curve	R/W	
51H	t 01	Time setting for the first segment of the third curve	R/W	
52H-77H	C02-t20	Setting of the 2nd to 20th segments of the third curve	R/W	
78H	C 01	The set value of the first segment of the fourth curve	R/W	
79H	t 01	Time setting for the first segment of the fourth curve	R/W	
7AH-9FH	C02-t20	Setting of the 2nd to 20th segments of the 4th curve	R/W	
A0H	SPC	curve selection	R/W	
A1H	CYC	Curve cycle times	R/W	

A2H	Sn	Enter specifications	R/W	
A3H	dP	decimal place	R/W	
A4H	dIL	lower limit display value	R/W	
A5H	dIH	upper limit display value	R/W	
A6H	FiLt	digital filtering	R/W	
A7H	ScL	Correction of lower limit of measurement value	R/W	
A8H	ScH	Backup	R/W	
A9H	Ctrl	control mode	R/W	
AAH	oPt	Output method	R/W	
ABH	dFct	Hysteresis value of ON-OFF control	R/W	
ACH	SF	Overshoot suppression coefficient	R/W	
ADH	oPL	Output Lower Limit	R/W	
AEH	oPH	Output upper limit	R/W	
AFH	P	Proportional band	R/W	
B0H	I	Integral time	R/W	
B1H	d	Differential time	R/W	
B2H	t	Output cycle	R/W	
B3H	Alt	Alarm method	R/W	
B4H	AL1	First alarm set value	R/W	
B5H	AL2	Second alarm set value (backup)	R/W	
B6H	AL3	Third alarm set value	R/W	
B7H	AL4	Preparation scope Of rdY function	R/W	
B8H	ALP	Alarm output port	R/W	
B9H	ALr	Alarm suppression method	R/W	
BAH	dF1	Difference value of switching of the first alarm	R/W	
BBH	dF2	Backup	R/W	
BCH	dF3	Backup	R/W	
BDH	dF4	Backup	R/W	

BEH	Sdt	Backup	R/W	
BFH	Addr	Communication address	R/W	
C0H	bAud	Communication baud rate	R/W	
C1H	CF	system function	R/W	
C2H	run	Run Settings	R/W	
C3H	Loc	Keyboard lock	R/W	
C4H		HMI Control Register 1	R/W	The Ones place meanings are as follows: Ones place bit write 1: Switch from stop to run state Ones place bit write 2: Switch to stop state Ones place bit write 3: Switch from running to paused state Ones place bit write 4: Switch from pause to running state Tens place bit write 1/2 corresponds to start/stop self-tuning Hundreds place bit write 1/2 corresponds to Switch to manual/automatic control mode
C5H		HMI Control Register 2	R/W	HMI Modifying the StEP value can directly write a new StEP value to the register
C6H		Control output value	R/W	After switching the instrument to manual control mode, this parameter can be modified to achieve HMI control output
C7H	LC	Remaining number of cycles	R/W	
C8H		Alarm output status	R	Bit0/bit1/bit2/bit3 respectively indicate the port

				status of OUT1/OUT2/OUT3/OUT4 used for alarm output, and=1 indicates the action of the output port
C9H	StEP	Current program segment	R	
CAH		Current working status	R	=0, the instrument is in a stopped state =1. The instrument is in operation =2. The instrument is in a paused state
CBH		Self tuning and manual control status	R	bit0=1, The instrument is in a self-tuning state bit1=1, The instrument is in manual control mode
CCH		Time already run	R	
CDH	PV	Current measurement value	R	
CEH	SV	Current setting value	R	

Explanation: 1) Communication data are all 16 bit signed integers, negative numbers are represented by complement codes, and the numerical range is -32768-32767.

2) Read/write properties: "R" is read-only, "R/W" is read-write, and writing to addresses that are not listed or do not have writable properties is prohibited.